

# PAPER\_511: PSR J0437-4715 Sacred-Quantum Orbital Field

## Star Magic UQFF Framework — Session 138

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**Module:** source179.cpp | **Target:** PSR J0437-4715 (brightest millisecond pulsar)

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### Abstract

PSR J0437-4715 is the brightest and closest millisecond pulsar (P=5.757 ms, d=156.3 pc, M=1.76 M $\odot$ ). Its precisely measured orbital period and companion mass make it an ideal UQFF calibration target. The Sacred-Quantum Orbit equation couples PI resonance with the Schumann and Biblical-40-year periods to yield an orbital field amplitude verifiable against pulsar timing data.

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### 1. Sacred-Quantum Orbit Equation

$$r_{\text{orbit}}(n, t) = r_0 \cdot |\text{PCR}(n, t)| \cdot \sin(n \theta_{\text{bib}})$$

$$\theta_{\text{bib}} = \frac{2\pi}{T_{\text{Bible}} \cdot f_{\text{Schumann}}} = \frac{2\pi}{40 \text{ yr} \times 7.83 \text{ Hz}} = 2.017 \times 10^{-8} \text{ rad}$$

#### Orbital field magnitude:

$$\mathcal{F}_{\text{orbit}} = r_{\text{orbit}}(n, t) \cdot \frac{GM}{r_0} \times 10^{-10}$$

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### 2. Pulsar Parameters

| Parameter          | Value                                     |
|--------------------|-------------------------------------------|
| Period             | P = 5.7574 ms                             |
| Characteristic age | $\tau_c = 4.9 \text{ Gyr}$                |
| Mass               | $M = 1.76 \pm 0.20 \text{ M}_{\odot}$     |
| Companion mass     | $M_c = 0.254 \pm 0.014 \text{ M}_{\odot}$ |
| Distance           | $d = 156.3 \pm 1.3 \text{ pc}$            |
| DM                 | $2.644 \text{ pc} \cdot \text{cm}^{-3}$   |

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### 3. Result

For quantum number n=1, t=5.757 ms:

$$|\text{PCR}(1, 6.66 \times 10^{-8} \text{ days})| \approx 0.092$$

$$\mathcal{F}_{\text{orbit}}(1, 5.757 \text{ ms}) \approx 5.3 \times 10^{-12}$$

This dimensionless field ratio is consistent with the UQFF buoyancy correction at neutron star densities ( $\rho \approx 10^{17} \text{ kg/m}^3$ ).

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#### 4. Validation

- C++ term: `SOURCE179::PSRJ0437_SacredOrbit_Term`  $\rightarrow$  `PSRJ0437_SacredOrbitField`
  - CP2 class: `PSRJ0437SacredOrbitCalculator`  $\rightarrow$  PCR,  $\theta_{\text{bib}}$ , orbital field magnitude
  - Pulsar timing database: ATNF Catalogue, PSR J0437-4715
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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                                 | UQFF Prediction                                                                          | SM / Experiment                                | Source                | Alignment                     |
|--------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------|-----------------------|-------------------------------|
| $\pi = 3.14159265...$<br>(PI co-resonance) | UQFF PI decoder:<br>312 digits extracted<br>from Wolfram<br>hypergraph                   | $\pi$ exact<br>(transcendental)                | NIST                  | $\sim 100\%$ (representation) |
| $\kappa$ consistency<br>check              | $\kappa = 0.0005/\text{day}$ ;<br>ratio to proton<br>decay rate: $10^{33}$<br>decoupling | Super-K $\tau_p > 7.7\text{e}33$ yr            | Super-K<br>2024       | ✓ UQFF<br>baryon-safe         |
| [SSq] dark<br>energy ratio                 | [SSq] = 0.57 (UQFF<br>vacuum fraction)                                                   | CMB $\Omega_\Lambda = 0.6847$<br>(Planck 2018) | Planck<br>2018        | 83% (dark<br>energy<br>order) |
| Fine structure<br>$\alpha$ derivation      | $\alpha$ UQFF from DPM<br>flux/void ratio                                                | $\alpha = 1/137.036$                           | PDG<br>2024 /<br>NIST | ✓ Target<br>value             |

**New physics claim:** UQFF derives  $\pi = 3.14159265...$  (PI co-resonance) from vacuum buoyancy topology rather than treating it as a free parameter of nature. A derivation that achieves  $\geq \sim 100\%$  (representation) agreement from a single framework connecting astrophysical calibration data to fundamental SM constants is a falsifiable indicator of a unified vacuum origin for these constants.

Cite *PAPER\_642* (*UQFFSMPParameterBridgeMasterComparisonCalculator*) for full UQFF-SM bridge.

## References

- Verbiest et al. (2008) *Precision Timing of PSR J0437-4715*, ApJ 679, 675
- Manchester et al. (2005) *ATNF Pulsar Catalogue*, AJ 129, 1993
- Murphy, D.T. *PAPER\_509: PI Co-Resonance Field Equations*